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An artificial intelligence language model improves readability of burns first aid information☆

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ABSTRACT

Aims: This study aimed to assess the potential of using an artificial intelligence (AI) large language model to improve the readability of burns first aid information.

Methods: An AI language model (ChatGPT-3) was used to rewrite content from the top 50 English-language webpages containing burns first aid information to be understandable by an individual with the literacy level of an 11-year-old, as recommended by the American Medical Association and Health Education England. The assessment of readability was conducted using five validated tools.

Results: In their original form, only 4% of the patient education materials (PEMs) met the target readability level across all tools. The median grade was 6.9 (SD=1.1). One-sample one-tailed *t*-test revealed that this was not significantly below the target ($p = .31$). After AI-modification, 18% of PEMs reached the target level using all tools, with a median grade of 6 (SD=0.9), which was significantly below the target level ($p < .001$). Once rewritten using AI, paired t-test demonstrated that all readability scores improved significantly ($p < .001$).

Conclusion: Utilising an AI language model proved an effective and viable method for enhancing readability of burns first aid information.

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1. Introduction

Burns continue to be a widespread global health concern, with ample evidence demonstrating that effective high-quality first aid is crucial to improve the morbidity and mortality of these injuries [1,2]. Despite these findings, however, the general population's understanding and application of burns first aid remain low [3–5].

The majority of persons now prefer to access health information online [6,7]. Effective navigation of online health information is dependent on an individual's health literacy, which is defined by The World Health Organisation (WHO) as 'the personal characteristics and social resources necessary for individuals and services to make informed decisions about health' [8]. Health literacy has been stated to have a stronger correlation with ill health than education level, deprivation, age or ethnicity [9]. Although no consensus on the recommended reading difficulty level for online health information is present, the American Medical Association (AMA) and Health Education England (HEE) suggests it should be written at a level that is comprehensible by an individual with a sixth-grade (age 11) literacy level, or lower [10,11].

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Readability refers to the ease of understanding or comprehension attributed to the writing style. Various tools have been developed to quantify readability, thereby providing writers and researchers a means of determining the suitability of a resource for its intended audience [12,13]. Previous research found that readability of burns first aid information available online is written at a level above that which is recommended by the AMA and HEE [14–16].

In recent years, artificial intelligence (AI) large language models, such as ChatGPT, have greatly improved in their accuracy and have become more readily accessible to healthcare professionals. The objective of this study was to assess whether an AI language model can be used to improve the readability of online burns first aid information.

2. Methods

A recent study carried out by the author that assessed the readability, accountability, and quality of burns first aid information available online had identified the top 50 English language webpages containing burns first aid information intended for the general population [14]. The Google.com search engine was selected for this search because Google searches comprised 91.4% of the online search market share at the time [17]. Exclusion criteria included: literature from peer-reviewed journals, sources written for healthcare professionals, or webpages that contained exclusively audio-visual material. The readability data from this dataset, which also included information on accountability and quality, was used in the current study.

The readability of original webpage content was calculated by extracting text from each webpage and transferring it to a Microsoft Word document. The extracted text underwent editing based on well-established protocols used in readability assessment studies [18,19]. Elements unrelated to patient information were meticulously removed to ensure the accuracy of the readability score. This included removing webpage navigation, website URLs, disclaimers, advertisements, copyright information, acknowledgements, author information, citations, and references. Additionally, supplementary editing of non-textual elements and punctuation, such as paragraph breaks, colons, and bullet points, was performed as these factors have been known to influence readability formulas and potentially lead to over- or underestimation of the text's readability [20].

For this study, a freely available AI language model, ChatGPT-3 (Chat Generative Pre-trained Transformer), was used to rewrite the text extracted from the previously identified and analysed webpages containing burns first aid information. This AI language model was prompted to rewrite the text to be understandable by an individual with the literacy level of an 11-year-old, which is the target advised by the AMA and HEE for health information [10,11]. The prompt used was as follows: 'revise the provided patient education material to have a readability level comprehensible by an individual with the literacy level of an 11-year-old, while maintaining a tone suitable for an adult audience'. The final clarifying statement was included in the prompt, as prior to its addition the language model produced text with a childlike tone, which was deemed inappropriate for the dissemination of patient information.

Using five validated readability tools accessible through online software, the readability of the AI re-written content was calculated [21]. These tools consisted of the Flesch Reading Ease Score (FRES), Flesch-Kincaid Grade Level (FKGL), Gunning Fog Index (GFI), Coleman-Liau Index (CLI), and Simple Measure of Gobbledygook Index (SMOG). Each tool applies a mathematical formula to the provided text passage (Supplementary Table 1). The calculation considers various factors, including the number of sentences, mean number of words per sentence, mean number of syllables per word, and the count of complex words (words with more than three syllables) [12,13]. The FRES produces a score from 0 to 100, where higher scores indicate easier readability, while the other formulas (FKGL, GFI, CLI, and SMOG) provide a reading age as an American grade level. Table 1 provides details on the interpretation of each tool and the target score for this study. The methodology employed for assessing readability in this study follows similar approaches used in studies examining patient information and patient education materials [13,22].

2.1. *Data analysis*

FRES, FKGL, CLI, SMOG, GFI and median grade scores were described as mean values ± standard deviation. One-sample

Table 1 – Interpretation of each readability score.

Readability Tool	Interpretation	Target score
FRES	90-100: Very easy (Grade 5) 80-90: Easy (Grade 6) 70-80: Fairly easy (Grade 7) 60-70: Standard (Grade 8-9) 50-60: Fairly difficult (Grade 10-12) 30-50: Difficult (College) 0-30: Very difficult (College graduate)	≥ 80
FKGL	Years of education required to understand most of the text	≤ 6.9
GFI	Years of formal education that a person requires in order to easily understand the text on the first reading	≤ 6.9
CLI	Years of education required to understand most of the text	6
SMOG	Reading grade level necessary for a person to have reached in order to fully understand text	≤ 6.9

FRES = Flesch Reading Ease, FKGL = Flesch-Kincaid Grade Level, GFI = Gunning Fog Index, CLI = Coleman-Liau Index, and SMOG = Simple Measure of Gobbledygook.

one-tailed t-test, with Ha: $\mu > 80$ for FRES and Ha: $\mu < 7$ for FKGL, GFI, CLI, SMOG and Median grade was calculated to test if mean scores were at the target readability level. The paired t-test was used to compare the mean readability scores of the original and AI-rewritten content. Statistical significance (alpha) was set at 0.05. All analyses were performed in R (v4.3.1).

3. Results

The original and AI-revised text from fifty webpages containing burns first aid information aimed at the general population were analysed (Supplementary Table 2). Distribution of webpages by readability level, pre- and post-AI revision can be found in Table 2.

For the original webpages, two (4%) were judged to be at the target reading level using all tools. Median grade was found to be 6.9 (SD=1.1). One-sample one-tailed t-test revealed that this was not significantly below the target reading level ($p = 0.310$). Mean FKGL (5.9, $p < .0001$) and SMOG (6.1, $p < .0001$) were found to be at the target reading level. Mean FRES (73.5, $p = 1.00$), GFI (8.2, $p = 1.00$) and CLI (8.3, $p = 1.00$) were all found to be above the target readability level.

Once revised using the AI language model, all readability scores improved significantly ($p < .001$), as can be seen in Table 3. Nine (18%) of the AI-revised webpages were at the target level using all tools. Median grade improved to 6 (SD=0.9), which was significantly below the target level ($p < .0001$). For the AI-revised text, mean FKGL (4.9, $p < .0001$) and SMOG (5.3, $p < .0001$) were found to be at the target reading level. Mean FRES (82.1, $p = .007$), GFI (7.4, $p = .986$) and CLI (6.9, $p = .271$) of the AI-revised text were all found to still be above the target readability level.

Following review of each original webpage and the AI-revised text, there were no identifiable factual errors or inaccuracies, taking into account the accuracy of the content that it had been prompted to revise.

Table 2 – Distribution of webpages by readability level.

Readability score by tool	Original Text		AI-generated Text	
	Frequency (%)		Frequency (%)	
FRES				
Very Easy (90-100)	0	(0)	6	(12)
Easy (80-89)	8	(16)	25	(50)
Fairly easy (70-79)	27	(54)	18	(36)
Standard (60-69)	14	(28)	1	(2)
Fairly difficult (50-59)	1	(2)	0	(0)
FKGL				
Grade 6 or below	37	(74)	47	(94)
Grade 7	7	(14)	2	(4)
Grade 8-9	6	(12)	1	(2)
GFI				
Grade 6 or below	9	(18)	19	(38)
Grade 7	13	(26)	14	(28)
Grade 8-9	21	(42)	15	(30)
Grade 10-12	7	(14)	2	(4)
CLI				
Grade 6 or below	6	(12)	19	(38)
Grade 7	6	(12)	14	(28)
Grade 8-9	29	(58)	15	(30)
Grade 10-12	9	(18)	2	(4)
SMOG				
Grade 6 or below	41	(82)	49	(98)
Grade 7	3	(6)	1	(2)
Grade 8-9	6	(12)	0	(0)
Median Grade Score†				
Grade 6 or below	25	(50)	41	(82)
Grade 7	12	(24)	9	(18)
Grade 8-9	13	(26)	0	(0)

† Calculated from median of FKGL, GFI, CLI and SMOG.
FRES: Flesch Reading Ease, FKGL: Flesch-Kincaid Grade Level, GFI: Gunning Fog Index, CLI: Coleman-Liau Index, SMOG: Simple Measure of Gobbledygook.

4. Discussion

Previous studies have highlighted the inadequacy of the quality and comprehensibility of burns first aid information available online. This includes written information, audiovisual content, and smartphone applications [14–16,23–26]. This study has shown that by using a freely available and simple to use AI language model, authors can improve the readability of burns first aid information.

It is vital that authors of health information consider the health literacy levels within the general population. Globally, one in ten adults lack 'the most basic information processing skills considered necessary to succeed in today's world' [27]. Additionally, nearly half of European adults exhibit health literacy categorised as 'inadequate' or 'problematic' [28]. In the United Kingdom, 43% of adults lack sufficient literacy skills to effectively understand health information, and in the USA, this percentage rises substantially to 88% [29,30]. It is because of this that the AMA and HEE recommends that health information should be written at a level comprehensible to an individual with the literacy level of an 11-year-old, or lower [10,11]. Improving readability can be achieved by using shorter words and sentences, and avoiding complex language. A range of toolkits are readily online to support the simplification of scientific content into more user-friendly communication materials [31].

Recently, AI language models have seen a surge in interest by medical researchers, especially since the release of the freely available ChatGPT-3, and later versions [32]. Much discourse has been published relating to the potential applications of these tools in healthcare and research. Recent literature has revealed how ChatGPT can be used to write discharge summaries, simplify radiology reports and even support cancer screening [33,34]. Justifiably, however, concerns exist around the content that these AI language models can produce, with reports of plagiarised and fabricated content generated when used to support the writing of scientific

Table 3 – Mean readability scores.

Readability score tool	Original Text		AI-generated Text		Mean Difference (95% CI)		p value[§]
	Mean Score (SD)		Mean Score (SD)				
FRES	73.5	(6.9)	82.1	(6.1)	8.6	(7.1-10.0)	< .00001
FKGL	5.9	(1.3)	4.9	(1.3)	1.1	(0.8-1.3)	< .00001
GFI	8.2	(1.5)	7.4	(1.3)	0.8	(0.6-1.1)	< .00001
CLI	8.3	(1.3)	6.9	(0.9)	1.4	(1.0-1.7)	< .00001
SMOG	6.1	(1.1)	5.3	(0.9)	0.8	(0.5-1.0)	< .00001
Median Grade Score[†]	6.9	(1.1)	6.0	(0.9)	0.9	(0.7-1.1)	< .00001

§ p value derived using paired t-test.
† Calculated from median of FKGL, GFI, CLI and SMOG.
FRES: Flesch Reading Ease, FKGL: Flesch-Kincaid Grade Level, GFI: Gunning Fog Index, CLI: Coleman-Liau Index, SMOG: Simple Measure of Gobbledygook, SD: Standard deviation, CI: Confidence interval.

articles [32]. A commonly identified issue is that these AI language models will not consistently advise on their limitations. Moreover, because these large language models have been trained using a wealth of preexisting literature, the responses they generate can perpetuate biases and stereotypes that exist in this content [35].

Large language models, such as ChatGPT, use deep learning to imitate language in a convincing and human-like way. As such, their potential to improve communication in healthcare must not be overlooked. In a recent study, a language model was trained to rewrite text in a more empathic way, which resulted in improved communication in a peer-to-peer mental health support system [36]. However, when used to generate answers to common questions pertaining to urological conditions and to produce patient information for common hand surgery procedures, it has recently been found that ChatGPT will generate content with a readability level well above that which is recommended by the AMA and HEE [37,38].

Discourse has been limited examining the use of AI language models to improve the readability of written health information. Recently, Kirchner and colleagues assessed whether ChatGPT could improve the readability of 20 orthopaedic patient education materials published by several leading American orthopaedic institutions pertaining to spinal pathologies and common arthroplasty procedures [39]. The results of their proof-of-concept study mirrors those of this study. They demonstrated that an AI language model was able to improve the readability of orthopaedic surgery online patient education materials quickly and effectively whilst maintaining sufficient detail and accuracy [39].

Clearly, AI language models can help make health information of all forms, including information leaflets and websites, more comprehensible. Previously, if an author of health information wanted to update their content to improve readability, they had to rewrite it manually, with the associated costs and time investment. Now, the process can be simplified, with the author only needing to review the AI-revised content to ensure accuracy. It is worth noting that all the content generated in this study was accurate, at least when compared to the original first aid information the AI language model was prompted to revise. This suggests that the AI language model primarily used the given text to generate the specifics of the new first aid information, but that the language used to express this was influenced by the content it was trained with. Using AI language models in this way can help bridge the gap in health literacy disparities, which is especially important given that lower health literacy has been linked with a lack of burns first aid knowledge [40].

5. Limitations

This study assessed the ability of an AI language model to improve readability at one point in time, however, this is likely to continue to improve as AI language models progress through subsequent iterations. AI language models are constantly being improved as they are provided with further training materials. Indeed, since this study was conducted, ChatGPT 3.5 was released. This study primarily focused on improving the readability as assessed by readability formulas. This methodology did not assess the impact of improved readability on users' comprehension, retention, or practical application of the information, which are crucial factors in assessing the effectiveness of first aid instructions. Future research should assess whether the improved readability is associated with better understanding and effectiveness of first aid. This study included no formal assessment of the content of the generated information, as this was outside of the scope of the study design. However, future work should be conducted to assess whether quality of AI re-written health information is maintained. AI language models are known to perpetuate biases and stereotypes present in their training data. In this study, AI generated content was not assessed for the presence of stereotypes or biases and this represents an important avenue in future research pertaining to AI generated patient education materials. As with the original study, this research was limited to English-language burns first aid information.

6. Conclusion

The challenge of inadequate health literacy levels in the general population is a critical concern, as it directly impacts individuals' ability to access and comprehend crucial health information, including first aid information. This study has demonstrated that utilising an AI language model, such as ChatGPT, is a viable and effective method to simplify and

enhance the readability of burns first aid information available online, although currently, to achieve a reading level in concordance with the AMA & HEE target, it is likely to require an iterative process of prompts. By employing AI technology, authors of health information can quickly and efficiently adapt their content to meet recommended readability levels. While AI language models offer significant benefits in improving health literacy, it is essential to remain vigilant regarding potential limitations, such as the perpetuation of biases and stereotypes present in the training data. Ethical considerations and content accuracy must continue to be a priority when utilising AI in healthcare communication.

Ethical approval

None required.

Financial disclosure statement

None.

CRediT authorship contribution statement

Alexander J. Baldwin: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. AJB read and approved the final manuscript.

Declaration of Competing Interest

The authors have no conflicting interests to disclose.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.burns.2024.03.005.

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